

SimuCredit Decisioning Model

Model Development and Initial Validation Report

Document ID	SDM-2026-LR-01
Version	1.0
Date	April 02, 2026
Prepared By	Quantitative Modeling Group — Model Risk & Analytics
Classification	Internal — Model Risk Management
Regulatory Framework	SR 11-7 ECOA / Reg B OCC 2011-12 EEOC 4/5ths Rule

1. Executive Summary

This report documents the development and initial validation of the **SimuCredit Decisioning Model (SDM-2026-LR-01)**, a binary classification model designed to support credit approval decisions at the point of application. The model outputs a probability of loan denial for each applicant; a decision threshold of $p = 0.50$ converts this to a binary approve/deny outcome.

The model was developed under the institution’s Model Risk Management Policy and the supervisory guidance set forth in **SR 11-7** (*Guidance on Model Risk Management*, Federal Reserve, 2011). Independent validation is required prior to any production deployment.

1.1 Performance Summary

Metric	Train	Test	Status
AUC (ROC)	0.8100	0.8110	✔ Acceptable (> 0.70)
Gini Coefficient	0.6199	0.6220	—
KS Statistic	—	0.5046	✔ Acceptable (> 0.30)
Hosmer-Lemeshow p	—	0.2410	✔ Acceptable fit (p > 0.05)
K-Fold CV AUC (k=5)	0.8057 ± 0.0706	—	⚠ Unstable

1.2 Fair Lending Summary

Protected Attribute	AIR (Approval Rate Ratio)	EEOC Threshold	Status
Gender (Female vs. Male)	0.9382	≥ 0.80	✓ Pass
Race (Minority vs. Majority)	0.7200	≥ 0.80	✗ FAIL — Adverse Impact Detected

 **Critical Finding:** Race AIR falls below the EEOC 4/5ths threshold. Mitigation analysis is required before deployment. See Section 7.

2. Data Description

2.1 Dataset Overview

The dataset (*SimuCredit_v2*) contains **682 anonymized consumer credit applications** drawn from the institution’s originations portfolio (2021–2023). Records were extracted from the loan origination system (LOS) and processed through the institution’s standard data quality pipeline.

Property	Value
Total Observations	682
Model Features	7 (all numerical)
Protected Attributes	2 (Gender, Race — excluded from model inputs)
Approved (Status = 0)	528 (77.4%)
Denied (Status = 1)	154 (22.6%)
Training Set	545 observations (80%, stratified)
Test Set	137 observations (20%, stratified)
Observation Period	2021 Q1 – 2023 Q4
Missing Values	< 3% across all features (imputed — see §4.2)

2.2 Feature Definitions

#	Feature	Type	Observed Range	Business Definition
1	Mortgage	Continuous (USD)	[0, 600,000]	Outstanding mortgage balance at time of application
2	Balance	Continuous (USD)	[0, 50,000]	Total revolving credit balance across all open accounts
3	Amount Past Due	Continuous (USD)	[0, 5,000]	Total amount currently past due across all accounts
4	Delinquency	Ordinal {0,1,2,3}	0–3	Maximum delinquency cycle in the past 24 months
5	Inquiry	Count	0–5	Hard credit inquiries in the past 12 months
6	Open Trade	Count	0–14	Number of currently open trade lines
7	Utilization	Ratio [0,1]	0.00–1.00	Aggregate credit utilization (balance / credit limit)

2.3 Protected Attributes (Excluded from Model — Fairness Audit Only)

The following demographic fields are present in the dataset but are **explicitly excluded from all model training steps** per ECOA / Regulation B. They are retained in a separate dataframe used only for post-hoc adverse impact analysis.

Attribute	Encoding	Protected Group	Reference Group	Proportion
Gender	Binary 0/1	Female (1)	Male (0)	45.2% Female / 54.8% Male
Race	Binary 0/1	Minority (1)	Majority (0)	28.0% Minority / 72.0% Majority

2.4 Class Distribution and Imbalance

The dataset exhibits significant **class imbalance**, with denial events comprising only **22.6%** of all observations. This is consistent with the expected denial rate for a prime consumer credit portfolio.

Status Distribution:

Approved (0):

Denied (1):

Total:

528 (77.4%)

154 (22.6%)

682

Note for Validation Team: Class imbalance was handled via stratified splitting. No SMOTE or oversampling was applied. The default threshold of p=0.50 may be suboptimal — threshold sensitivity analysis is required during validation.

3. Regulatory and Governance Context

3.1 Applicable Regulatory Framework

Regulation	Requirement
SR 11-7 (Federal Reserve, 2011)	Primary model risk management framework. Mandates conceptual soundness review, ongoing monitoring, and independent validation for all models used in material decisions.
ECOA / Regulation B	Prohibits credit discrimination on the basis of race, color, religion, national origin, sex, marital status, or age. Adverse action notices required for all denied applications.
OCC 2011-12	Parallel OCC model risk guidance. Requires validation of all models used in credit decisioning, with documentation of limitations and assumptions.
EEOC 4/5ths Rule	Adverse Impact Ratio (AIR) ≥ 0.80 required for all protected groups. $AIR = (\text{Approval Rate of Protected Group}) / (\text{Approval Rate of Reference Group})$.

3.2 Model Risk Tier Classification

This model is classified as **Tier 1 (High Risk)** under the Model Risk Policy due to its direct, automated impact on credit decisions affecting protected classes.

Tier 1 requirements:

- Full independent validation before production deployment
- Annual model performance review with documented findings
- Quarterly PSI-based monitoring reports
- Fair lending audit at each model update or threshold change
- Model Risk Committee (MRC) sign-off required for deployment

4. Model Development Methodology

4.1 Algorithm Selection Rationale

Logistic Regression was selected after evaluating four candidate algorithms:

Algorithm	Train AUC	Test AUC	Interpretability	Selected
Logistic Regression	0.8100	0.8110	High (coefficient table)	✅ Champion
Random Forest	0.8400	0.8210	Low (black box)	Challenger
XGBoost	0.8500	0.8160	Low	Rejected
Decision Tree (depth=4)	0.7600	0.7510	High	Rejected (lower AUC)

Logistic Regression was selected based on:

1. **SR 11-7 interpretability requirement** — coefficients provide direct, auditable log-odds mapping
2. **Regulatory precedent** — industry-standard for consumer credit under ECOA
3. **Comparable performance** — AUC within acceptable range of the RF challenger

4.2 Preprocessing Pipeline

The following pipeline is applied identically to training data and all future scoring data. Parameters are fitted **exclusively on training data** and frozen before any test-set evaluation.

```
Pipeline:
Step 1: Missing Value Imputation
Method: SimpleImputer(strategy='median')
Applied: All 7 feature columns
Fit on: Training data only. Frozen for scoring.

Step 2: Feature Scaling
Method: StandardScaler() →  $x' = (x - \mu) / \sigma$ 
Applied: All 7 feature columns (post-imputation)
Fit on: Training data only. Frozen for scoring.

Step 3: Protected Attribute Removal
Gender and Race are dropped BEFORE model training.
They are retained in a separate protected_data.csv for fairness audit.
```

4.3 Train / Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size = 0.20,
    stratify = y,      # preserves 6.5% denial rate in both splits
    random_state = 42
)
# Training: 545 rows | Test: 137 rows
```

4.4 Feature Selection — RCIT Independence Screening

A Randomized Conditional Independence Test (RCIT) was applied to screen features for conditional dependence on the target variable (Status). All seven features were retained ($p < 0.05$).

Feature	RCIT p-value	Decision
Mortgage	0.0030	✓ Retained ($p < 0.05$)
Balance	0.0110	✓ Retained ($p < 0.05$)
Amount Past Due	0.0004	✓ Retained ($p < 0.05$)
Delinquency	0.0001	✓ Retained ($p < 0.05$)
Inquiry	0.0280	✓ Retained ($p < 0.05$)
Open Trade	0.0420	✓ Retained ($p < 0.05$)
Utilization	0.0001	✓ Retained ($p < 0.05$)

4.5 Model Specification — Logistic Regression

The model is specified as a binary logistic regression:

$$\log\left(\frac{P(\text{Denial})}{1 - P(\text{Denial})}\right) = \beta_0 + \sum_{j=1}^7 \beta_j \tilde{x}_j$$

where $\tilde{x}_j$ denotes the standardized (zero-mean, unit-variance) version of feature x_j , and

$$P(\text{Denial} \mid \mathbf{x}) = \sigma(\beta_0 + \boldsymbol{\beta}^{\top} \tilde{\mathbf{x}}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^{\top} \tilde{\mathbf{x}})}}$$

Estimated Coefficients (on standardized features, L2 regularization, C=1.0):

Feature	β (Std. Scale)	Approx. Std. Error	Odds Ratio (e^{β})	Direction
Mortgage	+0.0373	0.0036	1.0380	↑ Higher → more likely denied
Balance	+0.2752	0.0312	1.3167	↑ Higher → more likely denied
Amount Past Due	+0.7340	0.0713	2.0834	↑ Higher → more likely denied
Delinquency	+0.8104	0.1025	2.2488	↑ Higher → more likely denied
Inquiry	+0.5871	0.0716	1.7988	↑ Higher → more likely denied
Open Trade	-0.0880	0.0114	0.9157	↓ Higher → less likely denied
Utilization	+0.4454	0.0770	1.5612	↑ Higher → more likely denied
Intercept (β_0)	-1.5319	—	—	—

Interpretation example: An applicant one standard deviation above the mean Utilization increases their log-odds of denial by +0.4454, corresponding to an odds multiplier of 1.561×.

Hyperparameters:

Parameter	Value	Rationale
Penalty	L2 (Ridge)	Prevents overfitting; standard for logistic regression
C (inverse regularization)	1.0	Default; no strong prior on coefficient magnitude
Solver	lbfgs	Efficient for small-to-medium datasets; supports L2
Max iterations	1,000	Convergence confirmed (loss < 1e-4)
Class weight	None	No resampling; stratified split preserves class balance
Threshold	0.50	Default; sensitivity analysis recommended by validation team

5. Model Performance Evaluation

5.1 Discriminatory Power

All metrics computed on the held-out test set (n=137) unless stated.

Metric	Train Set	Test Set	Gap	Benchmark
AUC (area under ROC curve)	0.8100	0.8110	-0.0011	> 0.70 = acceptable
Gini Coefficient (2·AUC – 1)	0.6199	0.6220	-0.0021	> 0.40 = acceptable
KS Statistic	—	0.5046	—	> 0.30 = acceptable

Train/Test AUC gap of -0.0011 (-0.1%) is within acceptable range (< 5% relative gap). Validation team should independently verify using k-fold CV.

5.2 Calibration Metrics

Metric	Value	Interpretation
F1 Score	0.5098	Harmonic mean of precision and recall
Precision	0.6500	Of predicted denials, fraction that are true denials
Recall (Sensitivity)	0.4194	Of actual denials, fraction correctly flagged
Specificity	0.9340	Of actual approvals, fraction correctly approved

5.3 Confusion Matrix (Test Set, threshold = 0.50)

		PREDICTED	
		Approve	Deny
ACTUAL	Approve	99	7
	Deny	18	13
True Negative Rate (Specificity): 0.9340			
True Positive Rate (Sensitivity): 0.4194			

5.4 Hosmer-Lemeshow Goodness-of-Fit Test

The Hosmer-Lemeshow test partitions applicants into deciles by predicted probability and compares observed vs. expected denial counts:

$$\chi^2_{HL} = \sum_{g=1}^G \frac{(O_g - E_g)^2}{E_g(1 - E_g/n_g)}$$

Statistic	Value
χ^2 (Hosmer-Lemeshow)	10.3548
Degrees of Freedom	8 (G – 2 = 10 – 2)
p-value	0.2410
H ₀	No significant difference between observed and expected frequencies
Decision	Fail to reject H ₀ — model fit is acceptable

5.5 K-Fold Cross-Validation (k=5, Stratified)

Fold	AUC
Fold 1	0.8904
Fold 2	0.7048
Fold 3	0.7988
Fold 4	0.7573
Fold 5	0.8774
Mean	0.8057
Std Dev	0.0706
95% CI	[0.6675, 0.9440]

Standard deviation of 0.0706 indicates fold-to-fold instability ($\sigma \geq 0.05$) — investigate data heterogeneity.

6. Multicollinearity Analysis — Variance Inflation Factors

VIF is computed to detect problematic linear dependencies among predictors. High VIF inflates coefficient standard errors and reduces interpretability.

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination from regressing feature j on all other features.

Feature	VIF	Interpretation
Mortgage	1.013	✔ No concern
Balance	1.008	✔ No concern
Amount Past Due	1.010	✔ No concern
Delinquency	1.018	✔ No concern
Inquiry	1.004	✔ No concern
Open Trade	1.017	✔ No concern
Utilization	1.004	✔ No concern

All VIF values are within acceptable bounds (< 5). No multicollinearity remediation required.
Coefficient estimates are stable.

7. Fair Lending Analysis — Adverse Impact Assessment

7.1 Methodology

Adverse Impact Ratio (AIR) is computed per the EEOC Uniform Guidelines on Employee Selection Procedures (29 C.F.R. § 1607), adapted to credit decisioning per Interagency Fair Lending Examination Procedures (CFPB / OCC / FDIC, 2009):

$$\text{AIR} = \frac{P(\text{Approved} \mid \text{Protected Group})}{P(\text{Approved} \mid \text{Reference Group})}$$

A model has **adverse impact** if $\text{AIR} < 0.80$ for any protected group (the EEOC 4/5ths, or 80%, rule).

7.2 Adverse Impact Results

Protected Attribute	Protected Group	Reference Group	Approval Rate (Protected)	Approval Rate (Reference)	AIR	Pass?
Gender	Female	Male	0.746	0.795	0.9382	 Pass
Race	Minority	Majority	0.600	0.833	0.7200	 FAIL

7.3 Findings and Required Actions

Gender: $\text{AIR} = 0.9382 \geq 0.80$. No adverse impact at the current threshold. Continued monitoring required.

Race: $\text{AIR} = 0.7200 < 0.80$. **Adverse impact detected.** The following remediation steps are required:

1. Slicing fairness analysis to identify which feature segments drive the disparity
2. Threshold adjustment analysis — determine whether adjusting the decision threshold improves AIR without unacceptable accuracy loss
3. Feature binning mitigation — evaluate whether binning proxy features reduces disparate impact
4. If mitigation is insufficient: escalate to Model Risk Committee and Legal for pre-deployment regulatory review

Validation Requirement: The independent validation team must re-run this analysis on the rebuilt model, perform slicing fairness analysis by feature segment, and document all mitigation steps taken.

8. Known Limitations and Model Assumptions

#	Limitation	Risk	Required Mitigation
1	Class imbalance (denial rate 22.6%) may bias toward approval; threshold sensitivity untested	Medium	Threshold sensitivity analysis during validation
2	No temporal ordering in train/test split; model may not capture economic cycle effects	Medium	PSI-based drift monitoring, quarterly
3	Linear log-odds assumption of logistic regression may miss nonlinear effects	Low	Validated via HL test and residual analysis; Random Forest challenger provides nonlinear benchmark
4	Protected attribute proxy risk: Utilization and Balance may be correlated with race/gender	High	Slicing fairness analysis; evaluate fairness-aware preprocessing for future model version
5	Model trained on single institution portfolio; generalizability to other portfolios is unvalidated	Low	Out-of-scope for current validation; flag for any portfolio expansion

9. Model Artifacts and Reproducibility

9.1 Artifact Inventory

Artifact	Filename	Format	Location
This report	simucredit_model_dev.md	Markdown	/docs/

9.2 Environment and Dependencies

python	3.11.4
scikit-learn	1.3.2
pandas	2.1.4
numpy	1.26.2
scipy	1.11.4
modeva	0.9.1

9.3 Reproduction Instructions

The validation team should reconstruct the model independently from this specification:

```

# 1. Load data
import pandas as pd
df = pd.read_csv('SimuCredit_v2.csv')

# 2. Define features and target
FEATURES = ['Mortgage', 'Balance', 'Amount Past Due',
            'Delinquency', 'Inquiry', 'Open Trade', 'Utilization']
X = df[FEATURES]; y = df['Status']

# 3. Split (must match: test_size=0.20, stratify=y, random_state=42)
from sklearn.model_selection import train_test_split
X_tr, X_te, y_tr, y_te = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=42)

# 4. Preprocess
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
imp = SimpleImputer(strategy='median').fit(X_tr)
scl = StandardScaler().fit(imp.transform(X_tr))
X_tr_s = scl.transform(imp.transform(X_tr))
X_te_s = scl.transform(imp.transform(X_te))

# 5. Train
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(C=1.0, max_iter=1000, random_state=42)
model.fit(X_tr_s, y_tr)

# 6. Evaluate — expected: AUC ≈ 0.8110, KS ≈ 0.5046
from sklearn.metrics import roc_auc_score
auc = roc_auc_score(y_te, model.predict_proba(X_te_s)[: ,1])

```

10. Validation Checklist and Sign-off

The following items are submitted to the independent validation team for review:

#	Validation Item	Status
1	Independent model reconstruction from this specification	Pending
2	Performance validation (AUC, KS, Gini, HL, K-Fold CV)	Pending
3	Robustness testing (feature perturbation, resilience)	Pending
4	Fairness audit — AIR for all protected attributes	Pending
5	Fairness mitigation analysis (if AIR < 0.80)	Pending
6	Interpretability review (coefficients, SHAP, PDP)	Pending
7	Challenger model comparison (LR vs. RF)	Pending
8	Validation report issuance	Pending
9	Model Risk Committee (MRC) review and approval	Pending

Role	Name	Signature	Date
Model Developer	[Redacted]	_____	April 02, 2026
Model Owner (Business Line)	[Redacted]	_____	April 02, 2026
Independent Validation Lead	<i>Pending</i>	_____	—
Chief Risk Officer	<i>Pending</i>	_____	—
MRC Chair	<i>Pending</i>	_____	—